

Recap

Availability is a very visible QoS attribute that refers to the readiness of services and systems

To ensure “good” availability levels, it is necessary to use reliable, i.e., robust and fault-tolerant, technological components with sufficient redundancies that avoid single point of failures

Trustworthiness is another aspect of dependability that refers to security and data privacy

DDoS attacks exploit DDoS for hire services to make attacks more effective (and more difficult to detect) and take down systems and services

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Types of DDoS attacks

DDoS attacks can be classified in various categories depending on their goal

1. *(Hyper-)Volumetric* or *flood-based* attacks designed to consume network bandwidth (implemented at the network, transport or application layers)
2. *Computational* attacks designed to consume CPU and memory (usually implemented at the application layer via TCP)
3. *Vulnerability-based* attacks designed to exploit software vulnerabilities (Zero-day, HTTP/2 Rapid Reset “request, cancel, request, cancel” pattern)

Attacks can also be classified as *symmetric* or *asymmetric* depending on the resources required on the attacker side

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Notable historical DDoS attack: Dyn

Massive cyberattack turned ordinary devices into weapons

by Samuel Burke @CNNTech

October 22, 2016: 10:37 AM ET



DDoS attack that disrupted internet was largest of its kind in history, experts say

Dyn, the victim of last week's denial of service attack, said it was orchestrated using a weapon called the Mirai botnet as the 'primary source of malicious attack'

- Major cyber attack disrupts internet service across Europe and US

Peak traffic 1Tbps

DDoS attack on Dyn came from 100,000 infected devices

DNS service provider Dyn says Mirai-powered botnets were the primary source for Friday's disruption



By Michael Kan

U.S. Correspondent, IDG News Service | OCT 26, 2016 2:21 PM PDT

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Other notable DDoS attacks

LILLY HAY, NEWMAN SECURITY 03.01.18 11:01 AM

GITHUB SURVIVED THE BIGGEST DDOS ATTACK EVER RECORDED

New world record DDoS attack hits 1.7Tbps days after landmark GitHub outage

Memcached denial-of-service attacks are getting bigger by the day, according to new analysis.

By Liam Tung | March 6, 2018 -- 12:34 GMT (12:34 GMT) | Topic: Security

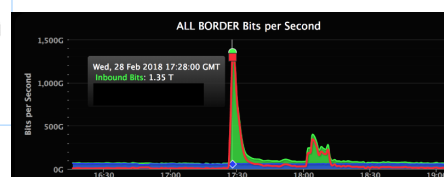
DDoS attacks exploited misconfigured *memcached* servers

Memcached is a high-performance open source distributed memory object caching technology designed to speedup dynamic web applications by alleviating database load

Amplification/reflection DDoS attacks coming from UDP port 11211 (i.e., spoofed requests to vulnerable servers that respond with a much larger amount of data, 50K amplification factor)

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Akamai Prevents the Largest DDoS Attack on a U.S. Financial Company

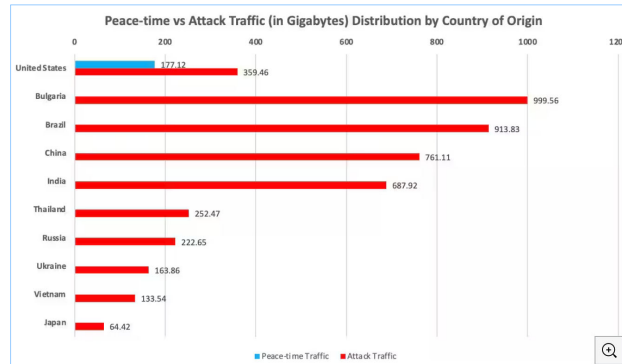


Craig Sparling & Sandeep Rath
September 08, 2023

Attack based on a combination of ACK, PUSH, RESET, and SYN flood attack vectors (peak: 633.7 Gbps and 55.1 Mpps)

It lasted two minutes

Cumulative traffic over 24 hours



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Largest DDoS attack (up to now)

Aisuru botnet sets new record with 31.4 Tbps DDoS attack

By Bill Toulas

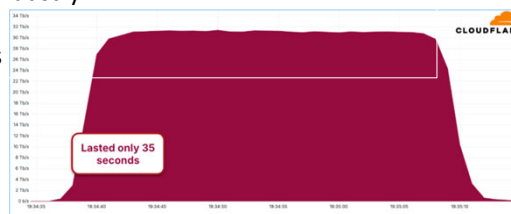
January 29, 2026

Infected Android TVs bombarded Cloudflare's network with hyper-volumetric HTTP DDoS attacks

Telcos emerged as the most-attacked industry

Traffic equivalent to 10,000 HD movies

World record: 31.4 Tbps DDoS attack autonomously mitigated by Cloudflare



Cloudflare just faced a self-inflicted DDoS-like outage from API bug it coded itself

Whoops!

By Christianna Silva on September 19, 2025



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DDoS summary

Impacts of DDoS:

- Unavailability of systems/services
- Financial loss
- Reputation loss

Mitigation actions depend on the type of systems/services under attack

- Disable UDP
- Prevent IP spoofing
- Use sophisticated (expensive) technologies

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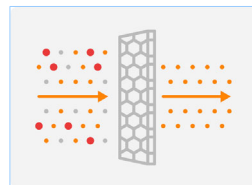
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Scrubbing centers

A centralized data cleansing solution aimed at mitigating DDoS attacks

Network traffic is inspected and proactive and AI-based mitigation controls designed to detect and stop attacks instantly are applied



Akamai provides a network of scrubbing centers

Traffic is usually redirected (using DNS or BGP) to the closest scrubbing center where potential attacks are mitigated (e.g., by blocking malicious IP addresses) and clean traffic is sent to the network for delivery

Akamai scrubbing centers are equipped to sustain high volumetric floods (more than 20 Tbps)

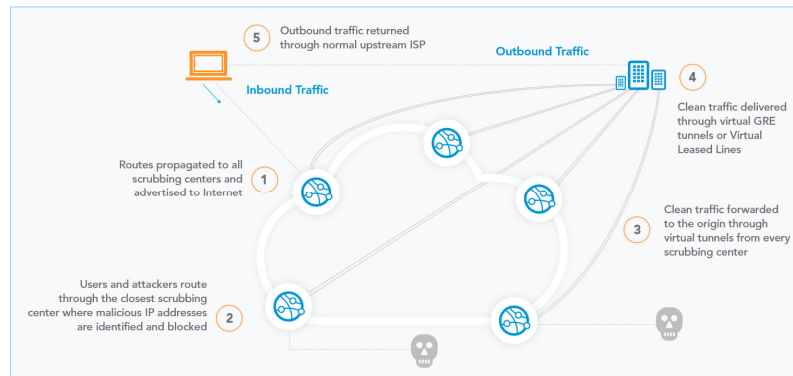
Often (and only) used by big enterprises, ISPs, cloud providers

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Akamai scrubbing architecture



Source: Akamai

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Data leakage/breach



Data leakage/breach refers to the *disclosure of sensitive data*

Three main categories of sensitive data that are targets of cyber-criminals:

- *Personally Identifiable Information (PII)* refers to data (e.g., name, home address, phone number, credit cards, health records) whose disclosure allows the identification of specific individuals (e.g., *identity theft*)
- *Business sensitive data* refer to data (e.g., intellectual property, designs, patents, financial data, customer/supplier info) whose disclosure to competitors poses a company at risk (e.g., *extortion*)
- *Classified information* refers to data (e.g., Wikileaks, Snowden & NSA) whose disclosure poses governments/countries at risk (e.g., *terrorism*)

Intentional data leakage: data captured as a consequence of a cyber attack

Unintentional data leakage: data disclosed accidentally to strangers by “owners”

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